

Sean Jackson

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Citizenship: US / UK / CA

EDUCATION

- **University of Toronto** Toronto, ON M5S 1A1
Honours BSc. Physics; Minors in Computer Science & Mathematics *September. 2022 – May. 2026*
 - Clubs: AI Club, Quantum Computing Club, Physics Student Union, Computer Science Student Union

WORK EXPERIENCE

- **Lawrence Livermore National Laboratory** Livermore, CA 94550, United States
Machine Learning Engineer (Materials Engineering Division, Summer Intern) *May. 2026 – Present*
 - **Materials Engineering:** Working within the Materials Engineering Division (MED) on machine learning applications for advanced materials research and manufacturing.
- **Lawrence Livermore National Laboratory** Livermore, CA 94550, United States
Machine Learning Engineer (Data Science Scholar Institute, Summer Intern) *May. 2025 – September. 2025*
 - **Project APEX:** Designed, trained, and deployed deep computer-vision models for high-throughput materials characterization in a [self-driving laboratory](#) for automated alloy discovery. Curated training datasets with Cornell University.
 - **Systems Design:** Built low-level robotics and tooling frameworks, integrated into high-level architectures for closed-loop automation (*details restricted under U.S. DOE regulations*).
 - **CV Design Challenge:** Designed generative decoding models for tracking occluded objects through walls using temporal feature representations.

PUBLICATIONS

- **Material Quality Classification Algorithms — First Author, Under Review (2025):** Novel end-to-end CV framework for automated materials surface quality assessment in self-driving labs. Led model design, implementation, and evaluation; imaging data from Cornell, validation with LLNL.

PROJECTS - [GITHUB](#)

- **GRACE (GRinding Automated Classification Engine)** [GitHub](#)
Custom PyTorch CV framework for automated materials quality assessment in self-driving labs. First-author paper under review.
- **SmartQueue** [GitHub](#) | [Website](#)
UofT office hours queue system with dynamic wait-time estimation, thematic clustering (DBScan + SBERT), and hybrid in-person/remote support.
- **Time Management Manager (TMM)** [GitHub](#)
Group time management system with dynamic multi-user availability and automated task scheduling.

SKILLS

- **ML & AI:** PyTorch, computer vision, architecture design, dataset engineering, model deployment (edge/limited-compute), RAG, LLMs, Transformers.
- **Quantum Computing:** Qiskit, Cirq, quantum algorithms, quantum information theory, quantum key encryption.
- **Software & Infrastructure:** Python, C/C++, Java, Linux, SQL, Git, CI/CD (GitHub Actions, Jenkins), AWS, Docker, Arduino/Raspberry Pi, MATLAB.
- **Physics & Instrumentation:** Computational physics (PDEs, Monte Carlo, FFT), laboratory instrumentation (lasers, optics, oscilloscopes, lock-in amplifiers), semiconductor fabrication (sputtering, photolithography).